

PROJECT ENGINEERING AND MANAGEMENT

ENEE 301

Lecture : 3

Tutorial : 1

Practical : 0

Year : III

Part : I

Course Objectives:

The objective of this course is to provide concepts, principles, and practices of project management, focusing on project planning, analysis, scheduling, monitoring, and control. By the end of the course, students will be able to evaluate project viability and apply modern management tools such as Gantt charts, Program Evaluation and Review Technique (PERT), Critical Path Method (CPM), Agile, and Scrum for effective scheduling, monitoring, and resource optimization.

1 Basics of Project Management (4 hours)

- 1.1 Fundamentals
 - 1.1.1 Definition, objectives, evolution (Traditional to agile)
 - 1.1.2 Types of projects (Infrastructures, research and development)
 - 1.1.3 Project versus operations
- 1.2 Key elements
 - 1.2.1 Triple constraints (Scope, time, cost)
 - 1.2.2 Stakeholder management, risk, quality, communication
- 1.3 Project life cycle
 - 1.3.1 Initiation, planning, execution, monitoring, closure
 - 1.3.2 Role of project management body of knowledge, agile frameworks

2 Project Initiation and Planning (8 hours)

- 2.1 Project initiation
 - 2.1.1 Opportunity analysis, SWOT analysis, business case
 - 2.1.2 Project charter, scope statement (Smart goals)
- 2.2 Feasibility study
 - 2.2.1 Technical, financial, economic and environmental
 - 2.2.2 Cost estimation (Top-down versus bottom-up)
- 2.3 Project appraisal
 - 2.3.1 Payback period, NPV, IRR, social cost-benefit analysis
 - 2.3.2 Risk assessment (Qualitative and quantitative)

3 Project Scheduling and Resource Management (8 hours)

- 3.1 Scheduling techniques
 - 3.1.1 Work breakdown structure (WBS)

- 3.1.2 Gantt charts and milestone tracking
- 3.1.3 CPM and PERT
- 3.2 Resource management
 - 3.2.1 Resource allocation, leveling, and optimization
 - 3.2.2 Budgeting, material and human resource planning

4 Project Procurement and Contracts (6 hours)

- 4.1 Procurement strategies
 - 4.1.1 Make-or-buy decision
 - 4.1.2 Vendor selection
 - 4.1.3 Expression of interest, requests for proposal, bidding process, contract types (Fixed-price, cost-plus)
- 4.2 Supply chain and legal aspects
 - 4.2.1 Procurement in agile versus traditional projects
 - 4.2.2 Contract management, dispute resolution

5 Monitoring, Control and Information Systems (4 hours)

- 5.1 Project control
 - 5.1.1 Earned value managements (EVM)
 - 5.1.2 Key performance indicators (KPIs), dashboards, variance analysis
- 5.2 Information systems
 - 5.2.1 Software and tools
 - 5.2.2 Reporting and communication plans

6 Evaluation, Auditing and Impact (4 hours)

- 6.1 Project evaluation
 - 6.1.1 Post-mortem analysis, lessons learned
 - 6.1.2 Return on investment, social and environmental impact
- 6.2 Auditing: Compliance, quality audits, ISO standards

7 Project Closure (3 hours)

- 7.1 Closing process
 - 7.1.1 Handover, documentation, final reports
 - 7.1.2 Contract closure, resource release
- 7.2 Lessons learned: Best practices, failure analysis

Tutorial (15 hours)

- 1. Project initiation and planning
- 2. Project scheduling and resource management: WBS, PERT and CPM tools
- 3. Administration task and exercise contracts
- 4. Monitoring, control and information systems exercise
- 5. Evaluation, auditing and impact
- 6. Project closure

7. Practical component: Group project
 - Group work of students (4-5 members) on a real-world project simulation
 - Use of project management software
 - Final presentation and report submission (Gantt chart, risk log, budget plan)

Final Exam

The questions will cover all the chapters in the syllabus. The evaluation scheme will be as indicated in the table below:

Chapter	Hours	Marks distribution*
1	4	5
2	8	10
3	8	10
4	8	10
5	8	10
6	6	10
7	3	5
Total	45	60

* There may be minor deviation in marks distribution.

References

1. Project Management Institute. (2021). A guide to the project management body of knowledge (PMBOK® Guide). Project Management Institute.
2. Kerzner, H. (2025). Project management: A systems approach to planning, scheduling, and controlling. Wiley.
3. Microsoft Corporation. (2024). Project Professional 2024.
4. Oracle Corporation. (2024). Primavera P6 Enterprise Project Portfolio Management.

DIGITAL SIGNAL PROCESSING AND APPLICATION

ENEX 304

Lecture : 3
Tutorial : 1
Practical : 3/2

Year : III
Part : I

Course Objectives:

The objective of this course is to impart fundamental knowledge and techniques for processing discrete-time signals and systems, including analog-to-digital and digital-to-analog conversion and the relationship between continuous-time and discrete-time signals. It emphasizes the analysis, design, and implementation of Finite Impulse Response (FIR) and Infinite Impulse Response (IIR) digital filters with desired frequency responses. By the end of the course, students will be able to apply the Discrete Fourier transform (DFT) using Fast Fourier transform (FFT) algorithms for practical signal processing applications in electrical engineering.

1 Discrete Time Signals and Systems (6 hours)

- 1.1 Basic elements of digital signal processing
- 1.2 Need of digital signal processing over analog signal processing
- 1.3 Sampling of continuous time signal, spectral properties of sampled signal
- 1.4 Discrete time signal, basic signal types
- 1.5 Transformation of independent variable
- 1.6 Energy signal and power signal
- 1.7 Periodicity of discrete time signal
- 1.8 Discrete time Fourier transform and properties
- 1.9 Discrete time system properties
- 1.10 Linear time invariant (LTI) system, properties of LTI system

2 Z-transform (3 hours)

- 2.1 Definition, Convergence of Z-transform and region of convergence
- 2.2 Properties of Z-transform (Linearity, time shift, multiplication by exponential sequence, differentiation, time reversal, convolution, multiplication)
- 2.3 Inverse z-transform by long division and partial fraction expansion

3 Analysis of LTI System in Frequency Domain (5 hours)

- 3.1 Frequency response of LTI system, response to complex exponential
- 3.2 Linear constant co-efficient difference equation and corresponding system function
- 3.3 Relationship of frequency response to pole-zero of system
- 3.4 Linear phase of LTI system and its relationship to causality

- 4 Discrete Filter Structures (6 hours)**
- 4.1 FIR filter, structures for FIR filter (Direct form, cascade, lattice)
 - 4.2 IIR filter, structures for IIR filter (Direct form I, direct form II, cascade, lattice, lattice ladder)
 - 4.3 Limit cycles
- 5 FIR Filter Design (9 hours)**
- 5.1 Filter design by window method, commonly used windows (Rectangular window, Bartlett window, Hanning window, Hamming window, Blackman window)
 - 5.2 Filter design by Kaiser window
 - 5.3 Filter design using optimum approximation, Remez exchange algorithm
 - 5.4 Types of FIR filters (Type-1, Type-2, Type-3 and Type-4)
- 6 IIR Filter Design (9 hours)**
- 6.1 Filter design by impulse invariance method
 - 6.2 Filter design using bilinear transformation
 - 6.3 Design of digital low pass Butterworth filter
 - 6.4 Frequency transformation of lowpass IIR filters (Transformation of lowpass digital filter prototype to high-pass, band-pass and band-stop filters)
- 7 Discrete Fourier Transform (5 hours)**
- 7.1 Discrete Fourier transform (DFT) representation, properties of DFT (Linearity, time shift, frequency shift, conjugation and conjugate symmetry, duality, convolution, multiplication), circular convolution
 - 7.2 Fast Fourier transform (FFT) algorithm (Decimation in time algorithm, decimation in frequency algorithm)
 - 7.3 Computational complexity of FFT algorithm
- 8 Applications of Digital Signal Processing (2 hours)**
- 8.1 DSP application in power system monitoring and diagnostics (Fault detection, power quality analysis)
 - 8.2 DSP application in protection and control systems (Smart grid management)
 - 8.3 DSP application in reliability and maintenance

- Tutorial (15 hours)**
- 1. Problems related to the conversion of analog signal to discrete time systems
 - 2. Problems related to identify whether the given signal is periodic or non-periodic signal
 - 3. Numerical problems to determine whether the given system is linear or non-linear, time invariant or time varying, causal or non-causal, stable or unstable

4. Problems related with linear convolution
5. Numerical problems related to z-transform and inverse z-transform
6. Solutions of difference equations
7. Solve the numerical problems related to frequency response of the LTI system
8. Draw the direct form-I, direct form-II, cascaded and lattice structure for FIR and IIR filters
9. Design of FIR filters using different windowing techniques (Rectangular window, Bartlett window, Hanning window, Hamming window, Blackman window, Kaiser window)
10. Design of Low pass IIR filters using impulse invariance method and bilinear transformation
11. Problems related to the frequency transformation of lowpass IIR filters to high pass, band pass and band stop filters (Using transformation equations)
12. Numerical problems depicting examples of DIT-FFT and DIF-FFT

Practical

(22.5 hours)

1. Introduction to Continuous Time and Discrete Time Signals
2. Linear Convolution
3. Linear Time Invariant System (LTI), Cascade of second order systems
4. Design of IIR filter (Impulse Invariance and Bilinear Transformation)
5. Design of FIR filter using window method. Comparison of FIR filter for different windowing method
6. Introduction to FFT

Final Exam

The questions will cover all the chapters in the syllabus. The evaluation scheme will be as indicated in the table below:

Chapter	Hours	Marks distribution*
1	6	8
2	3	4
3	5	6
4	6	8
5	9	12
6	9	12
7, 8	7	10
Total	45	60

* There may be minor deviation in marks distribution.

References

1. Oppenheim, A.V., Schafer, R.W., Buck, J.R. (2009). Discrete-time signal processing. Pearson Education.
2. Proakis, J.G., Manolakis, D.G. (2021). Digital signal processing: Principles, algorithms, and applications. Pearson.

3. Ludeman, L. (2009). Fundamentals of digital signal processing. John Wiley & Sons.
4. Tan, L., Jiang, J. (2025). Digital signal processing: Fundamentals, applications, and deep learning. Academic Press.
5. Schilling, R.J., Harris, S.L. (2004). Fundamentals of digital signal processing using MATLAB. CL Engineering.
6. ElAli, T.S. (2011). Discrete systems and digital signal processing with MATLAB. CRC Press.
7. Kani, A.N. (2021). Digital signal processing. CBS Publishers & Distributors.

DIGITAL CONTROL SYSTEM

ENEE 304

Lecture : 3
Tutorial : 1
Practical : 3/2

Year : III

Part : I

Course Objectives:

The objective is to impart knowledge and understanding of digital control systems, emphasizing the differences between analog and digital control, the role of sampling and data conversion, and the modeling of dynamic systems using difference equations and Z-transform techniques. By the end of the course, students will be able to apply stability analysis methods such as Jury's test and root locus for discrete-time systems and design digital PID controllers for discrete-time control applications.

1 Introduction to Discrete Time Control System (7 hours)

- 1.1 Advantages of digital control system over analog
- 1.2 Applications of digital control system
- 1.3 Types of signals
- 1.4 Sampling processes
- 1.5 Principle features of discrete time control system
- 1.6 Types of sampling operations
- 1.7 Signal sampling, quantizing and coding
- 1.8 Data acquisition and data distribution system
- 1.9 Sample and hold circuit
- 1.10 Quantizing and quantization error

2 The Z Transform (8 hours)

- 2.1 Discrete-time signals
- 2.2 Fundamental of the Z-transforms
- 2.3 Region of convergence
- 2.4 Z-transforms of some elementary functions
- 2.5 Important properties and theorems of the Z-transform
- 2.6 The inverse Z-transform
- 2.7 Z-transform method for solving difference equations

3 Z-plane Analysis of Discrete –Time Control Systems (8 hours)

- 3.1 Impulse sampling and data hold circuit
- 3.2 Z-transform by the convolution integral
- 3.3 Reconstructing original signals from sampled signals
- 3.4 Pulse transfer function

- 3.5 Starred Laplace transform of the signal involving both ordinary and starred Laplace transforms
- 3.6 Discrete time equivalents of continuous time system
- 3.7 Discrete time equivalents of analog controllers
- 3.8 Realization of digital controllers and digital filters

4 Stability, Design and Compensation of Discrete Time Control System (13 hours)

- 4.1 Mapping between the S-plane and the Z-Plane
- 4.2 Stability analysis of closed- loop systems in the Z-plane
- 4.3 Methods for testing absolute stability
- 4.4 Transient response analysis
- 4.5 Steady state response analysis
- 4.6 Use of root locus and frequency domain concepts
- 4.7 Compensator design based on the root locus and bode plot method
- 4.8 PID controller design and selection of parameters for discrete time system
- 4.9 Phase lead and phase lag compensator design for discrete time system

5 State Space Analysis (9 hours)

- 5.1 Concept of the state space method
- 5.2 State-space representations of discrete-time systems
- 5.3 State transition matrix
- 5.4 Pulse transfer function matrix
- 5.5 Discretization of continuous time state space equations
- 5.6 Stability assessment from the discretized state space equations

Tutorial (15 hours)

1. Problems involving Z-transforms of some elementary functions, important properties and theorems of the Z-transform, the inverse Z-transform, Z-transform method for solving difference equations
2. Problems on Z-transform by the convolution integral, realization of digital controllers and digital filters, discrete time equivalents of continuous time system and digital controllers and digital filters
3. Problems related to stability and design and compensation of discrete time control system
4. Problems related to state space analysis

Practical (22.5 hours)

1. Temperature Study of relay type "ON-OFF" control system
2. Z - transform using Software tools
3. Stability analysis of closed-loop system in z-plane using software tools
4. Simulation study of closed loop control system using software tools
5. Position control system through analog interfacing

Final Exam

The questions will cover all the chapters in the syllabus. The evaluation scheme will be as indicated in the table below:

Chapter	Hours	Marks distribution*
1	7	8
2	8	12
3	8	12
4	13	16
5	9	12
Total	45	60

* There may be minor deviation in marks distribution.

References

1. Ogata, K. (1995). Discrete-time control systems (Latest Edition). Prentice Hall.
2. Phillips, C. L. (1984). Digital control system: Analysis and design (Latest Edition). Prentice Hall.
3. Phillips, C. L., Nagle, H. T., Chakraborty, A. (2014). Digital control system analysis and design. Pearson.
4. Vashisth, R., Singh, K. (2012). Digital control system. Galgotia Publications.
5. Fadali, M. S., Visioli, A. (2012). Digital control engineering: Analysis and design. Academic Press.

SWITCHGEAR AND PROTECTION

ENEE 305

Lecture : 4
Tutorial : 1
Practical : 3/2

Year : III

Part : I

Course Objectives:

The objective of this course is to impart a comprehensive understanding of switchgear and protection in power systems, including the principles, construction, and operation of devices such as current transformer, potential transformer, fuses, isolators, contactors, circuit breakers, and relays. It emphasizes practical skills for testing, analyzing, and implementing protection schemes in real-world power applications. By the end of the course, students will be able to apply these concepts to design, operate, and troubleshoot switchgear and protection systems.

1 Fundamental of Power System Protection (4 hours)

- 1.1 Need for protective schemes
- 1.2 Types, causes and effects of faults
- 1.3 Functions of basic elements of a protective system
- 1.4 Zones of protection, primary and backup protection
- 1.5 Essential qualities of protection

2 Fuses, Isolators, and Contactors (8 hours)

- 2.1 Fuses: Characteristics, fusing factor, types and classification, selection, applications, discrimination
- 2.2 Isolators: Construction, operating principle, classification, types and applications
- 2.3 Contactors: Constructional features, NO/NC contacts, classifications, operating principles, applications and control circuits

3 Circuit Breakers (14 hours)

- 3.1 Need, importance in power system protection
- 3.2 Arc extinction methods: High resistance interruption, low resistance interruption
- 3.3 Arc phenomena, restriking and recovery voltages
- 3.4 Low voltage (LV) circuit breakers: Construction, operating principle, arc extinction, merits and demerits, application of MCB, MCCB, RCCB, ACB
- 3.5 High voltage (HV) circuit breakers: construction, operating principle, merits and demerits, arc extinction and application area of ABCB, BOCB, MOCB, VCB, SF₆ circuit breaker

- 3.6 HVDC circuit breakers: Principle, applications, challenges in DC arc extinction
- 3.7 Auto-reclosure in circuit breakers: Importance in transmission lines, single-shot and multiple-shot reclosing
- 3.8 Testing of circuit breakers: Type tests, routine tests and duty cycle test

4 Earthing and Protection Against Over Voltage (6 hours)

- 4.1 Earthing: Definition, purpose, system earthing and body earthing
- 4.2 Methods of earthing, earthing resistance, earth loop impedance
- 4.3 Causes of overvoltage in power system, lightning phenomena
- 4.4 Overvoltage protection: Overhead earth wire, angle of protection, horn-gap and rod gap lightning arrestors, surge absorbers.
- 4.5 Neutral earthing: Isolated neutral, solid neutral, resistance earthing, reactance earthing, Peterson coil earthing

5 Instrument Transformers for Protection (4 hours)

- 5.1 Current transformers (CTs): Construction, types, operation, ratios, burden, polarity, errors, knee-point voltage, selection, applications
- 5.2 Potential transformers (PTs): Construction, types, operation, ratios, accuracy, applications

6 Relay and Protection Schemes (14 hours)

- 6.1 **Protective relay: Classification and selection**
- 6.2 **Distance relay: Impedance, reactance and mho**
- 6.3 Electromagnetic attraction relays: Principle and applications
- 6.4 Electromagnetic induction relays: Principle and applications
- 6.5 Universal relay torque equation
- 6.6 Overcurrent protection: Construction and operation of instantaneous overcurrent relay.
- 6.7 Inverse definite minimum time (IDMT) relays
- 6.8 Time graded versus Current graded protection schemes
- 6.9 Directional overcurrent relay: Construction, torque equation, applications.
- 6.10 Distance and carrier aided protection
- 6.11 Unit protection schemes (Differential protection), biased (Percentage) differential protection, voltage balance relay
- 6.12 Protection schemes: Generator protection, transformer protection, busbar protection, feeder protection
- 6.13 Earth fault protection: Restricted and unrestricted earth fault protection

7 Static and Numerical Relays (10 hours)

- 7.1 Concept of static relays versus electromagnetic relays
- 7.2 Advantages and limitations of static relays
- 7.3 Comparator techniques

- 7.4 Static relays for protection: Differential relays, directional relays, impedance relays, admittance relays, mho relays
- 7.5 Numerical relays
 - 7.5.1 Description and architecture of numerical relays
 - 7.5.2 Functional blocks: Analog input module, A/D conversion, microprocessor, output relays
 - 7.5.3 Relaying algorithms and applications
 - 7.5.4 Advantages of numerical and microprocessor-based relays
 - 7.5.5 Applications in power system protection

Tutorial

(15 hours)

1. Design based on fundamentals of power system protection
2. Design based on fuses, isolators, and contactors
3. Numerical related to circuit breakers
4. Numerical related to earthing and overvoltage protection
5. Numerical related to relays and protection schemes

Practical

(22.5 hours)

1. Perform polarity test, CT ratio test and draw magnetizing curve for a protective CT; Check knee point voltage
2. Test over current and short circuit device in an air circuit breaker for operation using primary injection set
3. Test an induction disc relay in over current in over current protection scheme for operating characteristics using secondary injection set
4. Test an induction disc relay in residual earth fault protection scheme for operating characteristics and setting using secondary injection set
5. Check connections on a biased differential protection scheme of transformer/feeder; Test the scheme for operation and setting values on internal faults using primary injection set
6. Measurement of soil resistivity and earth electrode resistance

Final Exam

The questions will cover all the chapters in the syllabus. The evaluation scheme will be as indicated in the table below:

Chapters	Hours	Marks distribution*
1	4	6
2	8	8
3	14	12
4	6	6
5	4	4
6	14	14
7	10	10
Total	60	60

* There may be minor deviation in marks distribution.

References

1. Rao, S. S. (1977). Switchgear protection and power systems (Latest Edition). Khanna Publishers.
2. Gupta, J. B. (2013). Switchgear & protection. S. K. Kataria & Sons.
3. Amazon
4. Ram, B., Vishwakarma, D. N. (2011). Power system protection and switchgear. McGraw-Hill Education.
5. Ravindranath, B., Chander, M. (2011). Power system protection and switchgear. New Age International Publishers.
6. Paithankar, Y. G., Bhide, S. R. (2022). Fundamentals of power system protection. PHI Learning Pvt. Ltd.
7. Ingole, A. (2018). Switchgear and protection. Pearson India.
8. Uppal, S. L. (1988). Electrical power (Latest Edition). Khanna Publishers.

ELECTRIC MACHINE DESIGN

ENEE 306

Lecture : 3

Tutorial : 0

Practical : 3

Year : III

Part : I

Course Objectives:

The objective of this course is to provide fundamental knowledge of electrical conducting, magnetic, and insulating materials used in various electric machines, as well as the principles of heating, cooling, and overall performance optimization. It emphasizes on design of transformers, stators, and rotors of induction and DC machines. By the end of the course, students will be able to apply these concepts to analyze, design, and evaluate the performance of different electric machines.

1 Materials Used in Electrical Equipment (5 hours)

- 1.1 Review of electrical conducting materials
 - 1.1.1 Various characteristics and comparison between conducting materials
 - 1.1.2 Materials of high conductivity and high resistivity
- 1.2 Magnetic materials
 - 1.2.1 Classification, characteristics and application of magnetic materials
 - 1.2.2 Materials for steady flux (Solid core materials), materials for pulsating fluxes (Laminated core material sheet)
 - 1.2.3 Special purpose alloys, hot rolled and cold rolled steel sheets, sintered powder core
 - 1.2.4 Magnetic materials used in transformers, DC machines and AC machines
- 1.3 Insulating materials
 - 1.3.1 Classification, characteristics, application
 - 1.3.2 Insulating materials for transformers, DC machines and AC machines, ceramics

2 Heating and Cooling of Machines (7 hours)

- 2.1 Review of heat transfer: Conduction, convection and radiation
- 2.2 Internal temperature (Hot spots and their calculations)
- 2.3 Temperature gradients in iron core
- 2.4 Temperature gradients in conductors placed in slots
- 2.5 Ventilation of electrical machine
 - 2.5.1 Types of enclosures, methods of cooling, schemes of ventilation
 - 2.5.2 Cooling of totally enclosed machines, cooling circuits, cooling systems

- 2.6 Temperature rise, heating time constant, final steady temperature rise, cooling time constant
- 2.7 Rating of electric machine based on temperature rise
- 2.8 Calculation of temperature rise in armature, field coils and commutators

3 Transformer Design (14 hours)

- 3.1 Types of transformer: Power transformer, distribution transformer, core type and shell type
- 3.2 Design approach
 - 3.2.1 Output equations (Single and three phase), volt per turn
 - 3.2.2 Choice and sizing of core: Square core, stepped and cruciform core
 - 3.2.3 Choice of flux density
 - 3.2.4 Design of winding and choice of current density
 - 3.2.5 Design of insulation
 - 3.2.6 Design of window and window space factor
 - 3.2.7 Design of yoke and limb
- 3.3 Calculation of operating characteristics from design data
 - 3.3.1 Resistance of winding, leakage reactance of winding in core type transformer, iron loss, copper loss, efficiency, regulation
- 3.4 Design of cooling system
 - 3.4.1 Temperature rise in plain walled tank, design of tank and tubes

4 Three Phase Induction Motor Design (10 hours)

- 4.1 Three phase induction motor
 - 4.1.1 Construction, core material
 - 4.1.2 Types of three phase stator winding: Mush winding, lap winding and wave winding
- 4.2 Design approach
 - 4.2.1 Output equation, choice of magnetic and electric loading
 - 4.2.2 Choice of stator winding, stator slots and insulation, stator teeth, stator core and stator stamping dimension
 - 4.2.3 Air gap length, rotor design (Squirrel cage and slip ring type)
 - 4.2.4 Leakage inductance, evaluation of equivalent circuit parameters and operating characteristics from design data

5 DC Machine Design (9 hours)

- 5.1 Armature winding
 - 5.1.1 Lap and wave winding
- 5.2 Design approach
 - 5.2.1 Output equation, choice of average gap density, choice of ampere conductors per meter
 - 5.2.2 Choice of number of poles in DC machine, pole proportions

- 5.2.3 Selection of length of air gap
- 5.2.4 Choice of armature windings, number of armature conductors, number of coils, number of armature slots, armature conductor selection
- 5.2.5 Design of commutator, design of brushes, design of field windings and compensating winding

Practical

(45 hours)

1. Detailed design of core type power transformer: Orthographic drawing of transformer including winding, tank and tubes
2. Detailed design of core type distribution transformer: Orthographic drawing of transformer including winding, tank and tubes
3. Detailed design of three phase induction motor: Drawing of three phase stator winding (Mush winding, lap winding and wave winding)
4. Detailed design of DC armature winding: Drawing of lap and wave winding used in DC machine armature

Final Exam

The questions will cover all the chapters in the syllabus. The evaluation scheme will be as indicated in the table below:

Chapter	Hours	Marks distribution*
1	5	8
2	7	9
3	14	19
4	10	14
5	9	10
Total	45	60

* There may be minor deviation in marks distribution.

References

1. Sawhney, A. K. (2015). A course in electrical machine design. Delhi: Dhanpat Rai & Sons.
2. Deshpande, M. V. (2009). Design and testing of electrical machines. Delhi: PHI Learning Private Limited.
3. Say, M. G. (1957). The performance and design of A.C. machines (Latest Edition). London: Sir Isaac Pitman & Sons.
4. Say, M. G., Taylor, E. O. (1986). Performance and design of D.C. machines (Latest Edition). London: Pitman Publishing.
5. Agarwal, R. K. (2014). Principles of electrical machine design. S. Delhi: K. Kataria & Sons.

ENGINEERING THERMODYNAMICS AND HEAT TRANSFER

ENME 310

Lecture : 3
Tutorial : 1
Practical : 3/2

Year : III
Part : I

Course Objectives:

The objective of this course is to provide concepts of energy conservation and transfer, focusing on the laws of thermodynamics, thermodynamic cycles, and heat transfer, and exploring their applications in electrical engineering.

- 1 **Basic Concepts** (4 hours)
 - 1.1 Definition and scope of engineering thermodynamics
 - 1.2 Value of energy to society
 - 1.3 Microscopic versus macroscopic viewpoint
 - 1.4 Concepts and definitions
 - 1.4.1 Surface system, surroundings, boundary and universe; closed systems, open systems, and isolated systems
 - 1.4.2 Thermodynamic properties: Intensive, extensive and specific properties
 - 1.4.3 Thermodynamic equilibrium
 - 1.4.4 Thermodynamic state
 - 1.4.5 Thermodynamic process, cyclic process, quasi-equilibrium process, reversible and irreversible process
 - 1.5 Specific volume, pressure and pressure measurement devices, temperature and temperature measurement
 - 1.6 Zeroth law of thermodynamics, equality of temperature
- 2 **Energy and Energy Transfer** (3 hours)
 - 2.1 Energy and its meaning
 - 2.2 Stored energy and transient energy; Total energy
 - 2.3 Energy transfer
 - 2.3.1 Heat transfer
 - 2.3.2 Work transfer
 - 2.4 Expressions for displacement work transfer
 - 2.5 Other examples of work: Electrical work and mechanical forms of work
- 3 **Properties of Common Substances** (6 hours)
 - 3.1 Pure substance and state postulate
 - 3.2 Ideal gas and ideal gas relations

- 3.3 Two phase (Liquid and vapor) systems
 - 3.3.1 Phase change processes: The $T - v$, $P - v$ and $P - T$ diagrams
 - 3.3.2 Subcooled liquid, saturated liquid, wet mixture, critical point, quality, moisture content, saturated vapor and superheated vapor
 - 3.3.3 Properties of two-phase mixtures
 - 3.3.4 Compressibility factor
- 3.4 Development of property data: Graphical data presentation (P - h , h - s and T - s diagrams) and tabular data presentation

4 First Law of Thermodynamics (6 hours)

- 4.1 First law of thermodynamics for control mass
- 4.2 First law of thermodynamics for control mass undergoing cyclic process
- 4.3 Internal energy, enthalpy and specific heats
- 4.4 First law of thermodynamics for control volume
- 4.5 Control volume analysis: Steady state analysis
- 4.6 Control volume steady state work and flow applications
- 4.7 First law of thermodynamics for an isolated system and PMM-I type

5 Second Law of Thermodynamics (6 hours)

- 5.1 Necessity of formulation of second law
- 5.2 Concepts and definitions
- 5.3 Two phase (Liquid and vapor) systems
 - 5.3.1 Thermal reservoir, heat engine, heat pump, refrigerator
 - 5.3.2 Reversible and irreversible processes
- 5.4 Kelvin-Planck and Clausius statements of the second law of thermodynamics and their equivalence
- 5.5 Carnot cycle and Carnot efficiency
- 5.6 Clausius inequality and entropy
- 5.7 Second law of thermodynamics for an isolated system
- 5.8 Entropy changes of an ideal gases, liquids and solids from Gibbs equation
- 5.9 Concepts of change in entropy of control mass and control volume
- 5.10 Isentropic processes for an ideal gases, liquids and solids
- 5.11 Isentropic efficiencies of steady state applications

6 Gas Power Cycles (8 hours)

- 6.1 Classification of cycles
- 6.2 Air standard assumptions
- 6.3 Brayton cycle
 - 6.3.1 Analysis of closed and open cycles
 - 6.3.2 Cycle with intercooling, reheating, and regeneration
 - 6.3.3 Gas turbine power plant layout and components
- 6.4 Internal combustion cycles
 - 6.4.1 Operation of four strokes engine

- 6.4.2 Air standard Otto cycle
- 6.4.3 Air standard diesel cycle
- 6.4.4 Air standard Otto and diesel cycle comparison and mean effective pressure
- 6.5 Diesel power plant
 - 6.5.1 Essential components
 - 6.5.2 Plant layout
 - 6.5.3 Performance of I.C. Engines

7 Vapor Power Cycles and Vapor Compression Refrigeration Cycles (8 hours)

- 7.1 Rankine cycle
 - 7.1.1 Ideal and actual cycle
 - 7.1.2 Reheat cycle
 - 7.1.3 Regenerative cycle
 - 7.1.4 Essential components of steam power plant
 - 7.1.5 Layout of steam power plant
- 7.2 Vapor compression refrigeration cycle and its performance measurement

8 Heat Transfer (8 hours)

- 8.1 Basic concepts and modes of heat transfer
- 8.2 One dimensional steady state heat conduction through a plane wall
- 8.3 Radial steady state heat conduction through a hollow cylinder
- 8.4 Heat flow through composite structures
 - 8.4.1 Composite plane wall
 - 8.4.2 Multilayer tubes
- 8.5 Critical radius of insulation for cylinders
- 8.6 Electrical analogy for thermal resistance
- 8.7 Combined heat transfers and overall heat transfer coefficient for plane and composite wall and tube
- 8.8 Nature of convection; Free and forced convection
- 8.9 Heat transfer from extended surfaces; Fins, types of fins, fin equation, fin performance, fins effectiveness, proper length of fin
- 8.10 Heat radiation, Stefan's law, emissivity, absorptivity, reflectivity and transmissivity; Black body, white body and gray body

Tutorial (15 hours)

- 1. Sample problems related to the different thermodynamic properties
- 2. Sample problems related to displacement work transfer
- 3. Sample problems related to properties and work transfer of liquid-vapor mixture system (H₂O only)
- 4. Sample problems related to mass and energy conservation of control mass, cyclic processes and steady state control volume systems
- 5. Sample problems related to entropy change and entropy generation for ideal gas, isolated system and isentropic efficiencies

- Sample problems related to Otto, diesel and Brayton cycles and performance of I.C. engines
- Sample problems related to Rankine cycle
- Sample problems related to heat transfer from composite wall, cylinders, critical radius of insulation; Combined heat transfer, overall heat transfer coefficient

Practical

(22.5 hours)

- Temperature measurement by using different types of thermometers
- Experiment related to first law of thermodynamics
- Heat Pump: Coefficient of performance of heat pump/refrigerator
- Heat Conduction: Investigate Fourier's law of linear heat conduction
- Heat Radiation: Investigate Stefan-Boltzmann relationship
- Effect of extended surfaces on heat transfer

Final Exam

The questions will cover all the chapters in the syllabus. The evaluation scheme will be as indicated in the table below:

Chapters	Hours	Marks distribution*
1 and 2	7	10
3	6	8
4	6	8
5	6	8
6 and 7	12	16
8	8	10
Total	45	60

* There may be minor deviation in marks distribution.

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POWER ELECTRONICS

ENEE 302

Lecture : 3
Tutorial : 1
Practical : 3/2

Year : III

Part : I

Course Objectives:

The objective of this course is to provide concepts on various power electronic devices and their applications in power circuits. By the end of the course, students will be able to analyze and apply the operation and mathematical principles of different power electronic circuits.

1 Introduction (2 hours)

- 1.1 Concept of power electronics
- 1.2 Applications of power electronics
- 1.3 Advantages and disadvantages of power electronics converters
- 1.4 Introduction to power electronic system

2 Power Electronic Devices (6 hours)

- 2.1 Power diode
 - 2.1.1 Construction and operating characteristics
 - 2.1.2 Shockley diode equation
 - 2.1.3 Turn on transient and turn off transient
- 2.2 Thyristor
 - 2.2.1 Construction and operating characteristics
 - 2.2.2 Turn on characteristics
 - 2.2.3 Turn off process - natural commutation and forced commutation
 - 2.2.4 di/dt and dv/dt protections
 - 2.2.5 Thyristor firing circuits-with isolation transformer, pulse train generator, opto-coupler
 - 2.2.6 Advantages, disadvantages and applications of thyristors power diode
- 2.3 TRIAC
 - 2.3.1 Construction and operating characteristics
 - 2.3.2 Advantages, disadvantages and applications of thyristors
- 2.4 Gate turn off thyristor (GTO)
 - 2.4.1 Construction and operating characteristics
 - 2.4.2 Advantages, disadvantages and applications of GTO
- 2.5 Bi-polar junction transistor (BJT)
 - 2.5.1 Construction, operating characteristics and operation as a switch
 - 2.5.2 Base drive circuits

- 2.5.3 Advantages, disadvantages and applications of BJT
- 2.6 Metal oxide semi-conductor field effect transistor (MOSFET)
 - 2.6.1 Construction, operating characteristics and operation as a switch
 - 2.6.2 Advantages, disadvantages and applications of MOSFET
- 2.7 Insulated gate bipolar transistor (IGBT)
 - 2.7.1 Construction, operating characteristics and operation as a switch
 - 2.7.2 Advantages, disadvantages and applications of IGBT

3 Single Phase AC to DC Converters (10 hours)

- 3.1 Single phase half wave un-controlled rectifier with resistive load: Operating theory and waveforms, average and RMS values of output voltage, efficiency, ripple factor, Fourier analysis of output voltage
- 3.2 Single phase half wave un-controlled rectifier with inductive load: Operating theory and waveforms, average and RMS values of output voltage, load current equation
- 3.3 Single phase half wave rectifier with capacitor filter: Operating theory and waveforms, ripple factor
- 3.4 Single phase full wave un-controlled rectifier with center tap transformer: Operating theory and waveforms, average and RMS values of output voltage, efficiency, ripple factor, Fourier analysis of output voltage
- 3.5 Single phase full wave bridge rectifier with resistive load: Operating theory and waveforms, average and RMS values of output voltage.
- 3.6 Single phase full wave bridge rectifier with inductive load: Operating theory and waveforms, average and RMS values of output voltage, ripple factor, input current harmonics
- 3.7 Single phase half wave controlled rectifier with resistive load: Operating theory and waveforms, average and RMS values of output voltage, ripple factor, Fourier analysis of output voltage
- 3.8 Single phase full converter with resistive load: Operating theory and waveforms, average and RMS values of output voltage ripple factor, Fourier analysis of output voltage
- 3.9 Single phase full converter with highly inductive load: Operating theory and waveforms, average and RMS values of output voltage ripple factor, input current harmonics, Fourier analysis of output voltage and input ac current, input displacement factor, distortion factor, input harmonic factor, total harmonics distortion (THD), input power factor
- 3.10 Input power factor improvement: Extinction angle control, symmetrical angle control, multiple pulse width control

4 Three Phase AC to DC Converters (6 hours)

- 4.1 Three phase single way un-controlled rectifier with resistive load: Operating theory and waveforms, average and RMS values of output voltage, ripple factor, Fourier analysis of output voltage

- 4.2 First three phase un-controlled bridge rectifier with resistive load: Operating theory and waveforms, average and RMS values of output voltage, ripple factor, Fourier analysis of output voltage
- 4.3 Three phase single way controlled rectifier with resistive load: Operating theory and waveforms, average and RMS values of output voltage, ripple factor, Fourier analysis of output voltage
- 4.4 Three phase controlled bridge rectifier with resistive load: Operating theory and waveforms, average and RMS values of output voltage, ripple factor
- 4.5 Three phase full converter: Operating theory and waveforms, average and RMS values of output voltage
- 4.6 Twelve pulses of operation of three-phase full converter
- 4.7 Series connection of full converter

5 DC Chopper

(6 hours)

- 5.1 Step down DC chopper with resistive load: Operating theory and waveforms, average and RMS values of output voltage, output power, effective input resistance, constant frequency mode of operation, variable frequency mode of operation
- 5.2 Step down dc chopper with DC motor load: Operating theory and waveforms, load current equation, average load current
- 5.3 Step up chopper with resistive load
- 5.4 Buck-boost converter
- 5.5 Bi-directional dc to dc converter: Operating principle, step-up and step-down operation, application in electric vehicle
- 5.6 Chopper classification: Type-A, Type-B, Type-C, Type-D, Type-E

6 Inverters

(10 hours)

- 6.1 Single phase inverter with square wave AC output and resistive load
 - 6.1.1 Operating theory and waveforms, RMS value and AC motorload
 - 6.1.2 Fourier analysis of output voltage
- 6.2 Single phase inverter with square wave AC output and AC motor load
- 6.3 Three-phase inverter
 - 6.3.1 Three-phase inverter with 180° conduction mode with six-steps square wave output and resistive load
 - 6.3.2 Three-phase inverter with 120° conduction mode with six-steps square wave output and resistive load
 - 6.3.3 Fourier analysis of 3-phase inverter output voltage, positive, negative and zero sequence series
- 6.4 Pulse width modulated (PWM) inverter
 - 6.4.1 Single pulse width modulation
 - 6.4.2 Multiple pulse width modulation
 - 6.4.3 Sinusoidal pulse width modulation
- 6.5 Current source inverter

7 AC Voltage Controller

(5 hours)

- 7.1** Single phase AC voltage controller with resistive load: Operating theory and waveforms, RMS values of output voltage, output power, Fourier analysis of output voltage waveform
- 7.2** Single phase AC voltage controller with R-L inductive load: Operating theory and waveforms, load current equation RMS values of output voltage and load current
- 7.3 Single phase AC voltage controller with induction motor load: Operating theory and waveforms, load current equation RMS values of output voltage and load current
- 7.4 Three phase AC voltage controller with resistive load: Operating theory and waveforms, RMS values
- 7.5** Cyclo-converter
 - 7.5.1 Single phase step-down cyclo-converter
 - 7.5.2 Three phase to single phase step-down cyclo-converter
 - 7.5.3 Three phase to three phase step-down cyclo-converter

Tutorial

(15 hours)

- 1. Numerical related to Schockley diode equation and turn off transient
- 2. Numerical related to di/dt and dv/dt protection of thyristor and thyristor firing circuits
- 3. Numerical related to single phase half wave and full-wave uncontrolled rectifier with resistive load and inductive load
- 4. Computer programming to calculate Fourier coefficients output voltage waveform of single phase full wave rectifier up to 36th order and re-construction of wave-forms
- 5. Numerical related to single phase half wave and full-wave controlled rectifier with resistive load and inductive load
- 6. Numerical related to capacitor filter for single phase un-controlled rectifier
- 7. Numerical related to extinction angle control, symmetrical angle control, multiple pulse width control
- 8. Numerical related to three-phase un-controlled and controlled rectifier
- 9. Numerical related to DC chopper: Step up, step down and Buck-boost
- 10. Numerical related to single phase inverter with resistive load, ac motor load
- 11. Computer programming to calculate Fourier coefficients output voltage waveform of single phase inverter up to 36th order and re-construction of wave-forms
- 12. Numerical related to Pulse Width Modulated Inverters
- 13. Numerical related to single phase AC voltage controller with resistive and inductive loads
- 14. Computer programming to calculate Fourier coefficients output voltage waveform of single phase AC voltage controller up to 36th order and re-construction of wave-forms
- 15. Numerical related to cyclo-converter

Practical**(22.5 hours)**

1. Study of single phase half wave and full wave uncontrolled rectifier circuit and effect of capacitor filter
2. Study of single phase half-wave controlled rectifier and single phase full-wave controlled rectifier circuits
3. Study of three-phase single way un-controlled and controlled rectifier
4. Study of step down DC-DC chopper
5. Study of single phase AC voltage Controller
6. Study of three-phase inverter with resistive load
7. Simulation study of grid-connected 3-phase inverter with hysteresis band current control

Final Exam

The questions will cover all the chapters in the syllabus. The evaluation scheme will be as indicated in the table below:

Chapter	Hours	Marks distribution*
1	2	4
2	6	6
3	10	12
4	6	8
5	6	10
6	10	12
7	5	8
Total	45	60

* There may be minor deviation in marks distribution.

References

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